

EXERCISE 2

1. SELECT DISTINCT department

FROM students;

students

student-id	Department
1	IT
2	HR
3	IT
4	Finance
5	HR

2. SELECT department, AVG(age) AS avg-age

FROM students

GROUP BY department;

students

id	department	Avg-age
1	Alice IT	20
3	Charlie IT	21
2	Bob HR	22
4	Eve HR	22
5	Diana Finance	23

3. SELECT department, COUNT(*) AS student-count

FROM students

GROUP BY department

HAVING COUNT(*) > 1;

students

id	department	student-count
	IT	2
	HR	2

4. SELECT student-id, name, age, department

FROM students

WHERE age BETWEEN 21 AND 23;

students

student-id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. SELECT student-id, name, age, department

FROM students

WHERE department IN ('IT', 'HR')

AND age > 21;

students

student-id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

6. SELECT department, SUM(credits) AS total-credits

FROM courses

GROUP BY department

HAVING SUM(credits) > 5;

courses

department	total-credits
IT	10

7. SELECT course-id, course-name, department, credits
FROM courses
WHERE credits < 4;

courses			
course-id	course-name	department	credits
104	Excel	Finance	2
105	Statistics	HR	3

8. SELECT course-id, course-name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;

Courses		
course-id	course-name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3
105	Statistics	3

9. SELECT MAX(grade) AS max-grade,
MIN(grade) AS min-grade,
AVG(grade) AS avg-grade
FROM enrollments;

enrollments		
max-grade	min-grade	avg-grade
90	78	85

10. SELECT course-id, COUNT(*) AS enrollment-count
FROM enrollments
GROUP BY course-id;

enrollments	
course-id	enrollment-count
1	5

11. SELECT department,
SUM(salary) AS total-salary,
SUM(bonus) AS total-bonus
FROM salaries
GROUP BY department;

salaries		
department	total-salary	total-bonus
IT	100 000	10 500
HR	109 000	7500
Finance	70 000	6000

12. SELECT department, AVG(salary) AS avg-salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55 000;

salaries	
department	avg-salary
IT	100 000
Finance	70 000

13. SELECT employee_id,
 name,
 salary,
 bonus,
 (salary+bonus) AS total-compensation
 FROM salaries
 WHERE (salary+bonus) > 60 000;

Salaries				
employee-id	name	salary	bonus	total-compensation
1	Tom	60 000	5000	65 000
3	Spike	70 000	6000	76 000
4	Tyke	60 000	5500	67 500

14. SELECT department,
 SUM(budget) AS total-budget,
 AVG(budget) AS avg-budget
 FROM projects
 GROUP BY department
 HAVING AVG(budget) > 70 000;

Projects		
department	total budget	avg. budget
IT	270 000	¹²² 300 000

15. SELECT project-id, project-name, department, budget
 FROM projects
 WHERE budget BETWEEN 50 000 AND 120 000
 AND department <> 'Marketing';

Projects			
project-id	project-name	department	budget
5	HR Portal		50 000
2	Payroll System		80 000
1	AI App		120 000
3	Dash board		150 000